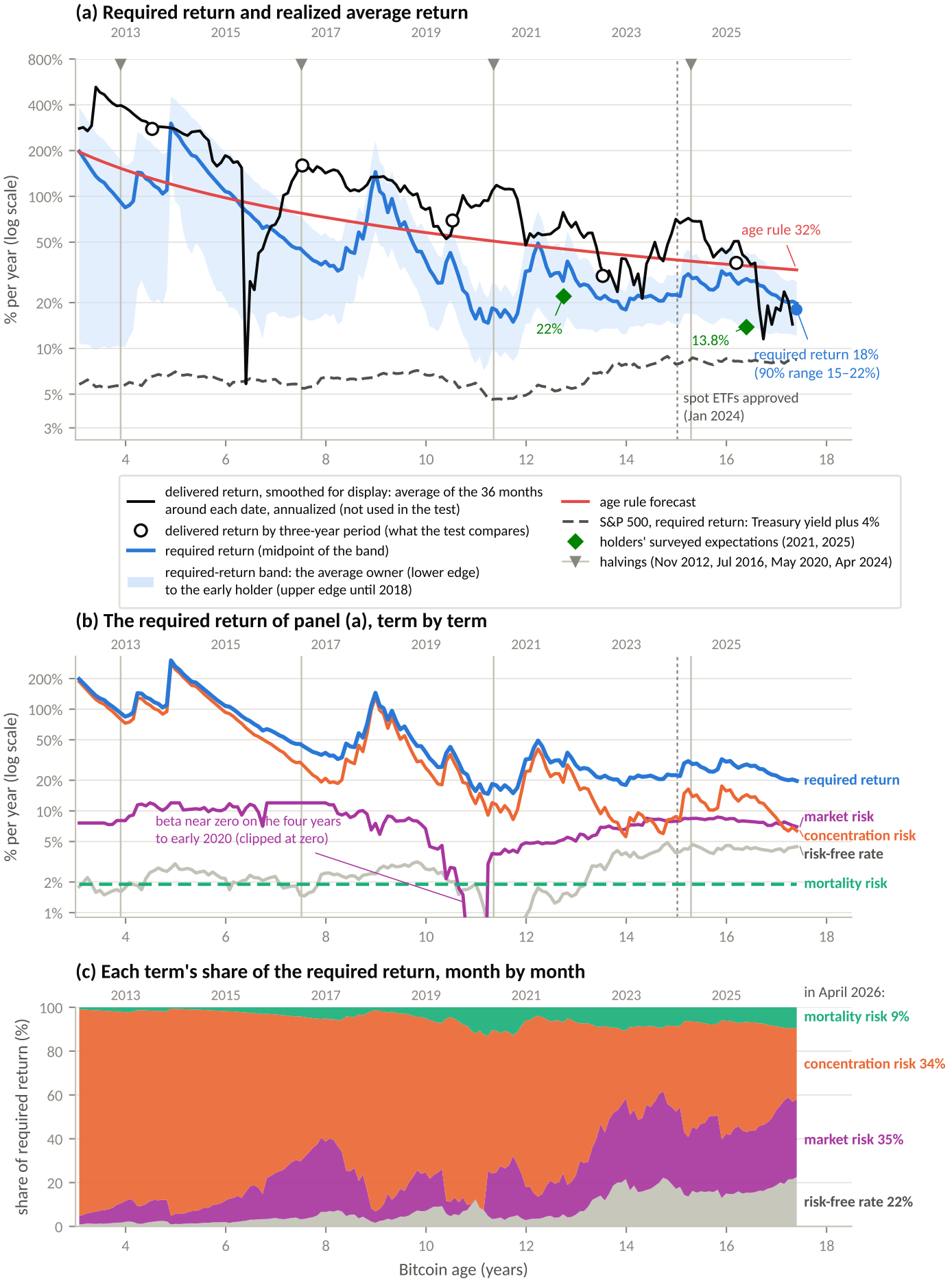

(a) Required return and realized average return
2013
2015
2017
2019
2021
2023
2025
800%
400%
200%
100%
50%
20%
10%
5%
3%
% per year (log scale)
age rule 32%
22%
13.8%
required return 18%
(90% range 15–22%)
spot ETFs approved
(Jan 2024)
4
6
8
10
12
14
16
18
delivered return, smoothed for display: average of the 36 months around each date, annualized (not used in the test)
delivered return by three-year period (what the test compares)
required return (midpoint of the band)
required-return band: the average owner (lower edge) to the early holder (upper edge until 2018)
age rule forecast
S&P 500, required return: Treasury yield plus 4%
holders' surveyed expectations (2021, 2025)
halvings (Nov 2012, Jul 2016, May 2020, Apr 2024)
(b) The required return of panel (a), term by term
2013
2015
2017
2019
2021
2023
2025
200%
100%
50%
20%
10%
5%
2%
1%
% per year (log scale)
beta near zero on the four years
to early 2020 (clipped at zero)
required return
market risk
concentration risk
risk-free rate
mortality risk
4
6
8
10
12
14
16
18
(c) Each term's share of the required return, month by month
2013
2015
2017
2019
2021
2023
2025
in April 2026:
mortality risk 9%
concentration risk 34%
market risk 35%
risk-free rate 22%
100
80
60
40
20
0
share of required return (%)
4
6
8
10
12
14
16
18
Bitcoin age (years)